DILLAN IMANS

Computer Vision Researcher

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PROFILE

M.Sc. student in AI in Biomedicine at TU Munich and computer vision researcher with over two years of experience in medical imaging and robot-assisted surgical settings. Growing focus on perception for robotics, and eager to build robotics software for medical applications.

EDUCATION

M.Sc. AI in Biomedicine, Technical University of Munich **Oct 2026 – Present**
Member of Elite Network of Bavaria, Imaging and Sensing focus

B.S. Computer Science & Engineering, Sungkyunkwan University **Aug 2022 - Jun 2026**
95.7 / 100 · Dean's List · Academic Excellence Scholarship

EXPERIENCE

Graduate Computer Vision Researcher @ CAMP, TUM **Oct 2026 - Present**

- Developing a self-supervised multimodal OR encoder that fuses multi-view video, 3D point clouds, speech/audio, and SAM2-based tracking and pose data for surgical scene understanding

Undergraduate Computer Vision Researcher @ Superint Lab, SKKU **Oct 2024 - Jun 2026**

- Built an end-to-end retinal screening system (optic disc/cup segmentation, vessel analysis) and its backend; ONNX-optimized for low-latency inference and delivered to a clinical partner
- Developed domain adaptation (+9.0 Dice across 6 BraTS 2021 settings, IEEE BIBM '25) and cross-modal distillation (up to +5.2 AUC, MICCAI workshop 2026) frameworks for medical image segmentation and classification

Undergraduate Computer Vision Researcher @ Labren, CUHK **Jun 2024 - Jun 2026**

- Curated data and ran initial experiments for a surgical robot policy learning paper via world modeling, with NVIDIA
- Built preprocessing infrastructure and an annotation QA pipeline for the first large-scale surgical action recognition dataset (11,915 clips)

PROJECTS

Haptic Needle Teleoperation @ HEX lab, TUM **Oct 2026 - Present**

- Teleoperation setup in which a Force Dimension haptic device drives a Franka robot arm carrying a needle, aimed at needle-insertion tasks (early stage)
- Adding multimodal operator feedback (haptic, vibration, audio, tongue-based tactile) so the operator can perceive what the needle is doing; working in ROS and C++

SELECTED PUBLICATIONS

Full list: scholar.google.com/citations?user=o-TtUq0AAAAJ · Reviewer, EMA4MICCAI 2026

1. **Imans, D.**, et al. Unsupervised Domain Adaptation with SAM-RefiSeR for Enhanced Brain Tumor Segmentation. *IEEE BIBM 2025*.
2. **Imans, D.**, et al. Clinical Graph-Mediated Distillation for Unpaired MRI-to-CFI Hypertension Prediction. *EMA4MICCAI 2026*.
3. Xu, M., ..., **Imans, D.**, et al. Generalized Recognition of Basic Surgical Actions Enables Skill Assessment and VLM-based Surgical Planning. *npj Digital Surgery (accepted)*.
4. He, Y., ..., **Imans, D.**, et al. Cosmos-H-Surgical: Learning Surgical Robot Policies from Videos via World Modeling. *Preprint*.

SKILLS & LANGUAGES

Programming: Python (advanced) · C++ (intermediate) · Java · TypeScript
Vision & ML: PyTorch · MONAI · nnU-Net · SimpleITK · OpenCV · pydicom · NiBabel · ONNX
Tooling: Git · GitHub Actions · Linux · Docker · Bash
Languages: English, Indonesian (fluent) · Mandarin, Korean (conversational) · German (B1)